End-Term Examination (Regular & Reappear)
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<ECE-AI/ ECE/ CSE>, Semester:<1>
(November-December, 2023)

Subject Code: BCS 110	**Subject: Programming in C Language**

Time :3 Hours **Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	
	(a) What is an operating system? Explain its function.		
	(b) Draw the flow chart of the following while and do while loop.		
	(c) Explain the concept of a break statement with the example.		
	(d) Write a program to convert a decimal number to an octal number		

	(e) What will be the count's value after executing the given function at fun(13)?
	`int fun (int num)`
	`{`
	`int count = 0;`
	`while (num)`
	`{`
	`count++;`
	`num >>= 1;`
	`}`
	`return (count);`
	`}`		

	(f) What is recursion? Explain with the help of an example		

	(g)
	Find the output of the following code.
	`#include<stdio.h>`
	`void f(int *p, int*q)`
	`{`
	`p=q;`
	`*p=2;`
	`}`
	`int i=0, j=1;`
	`int main()`
	`{`
	`f(&i, &j);`
	`printf("%d%d\n",i,j);`
	`return 0;`
	`}`		

	(h) What are the differences between call by value and call by reference? Explain with an example.		
	UNIT-I		
Q2	a) Explain the following in detail: ALU and register. b) Explain the following in detail: main memory and secondary memory.	**(5+5)**	
Q3	a) Write the flow chart of the switch case and if-else	**(10)**	

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	ladder statements. b) What are the differences between compiler and interpreter?		
	UNIT-II		
Q4	a) Explain the concept of any two logical operators with the help of an example. b) Explain getchar() and putchar() functions with example.	**(5+5)**	
Q5	a) What are the data types? Explain any two data types with examples. b) Write a c program to identify whether the given year is a leap or not.	**(5+5)**	
	UNIT-III		

Q6	a) Write a program for matrix multiplication in c such that the order of the first matrix is 3x3 and the order of the second matrix is also 3x3.
	b) Write a program to print the following pattern;
	`* * * * * * * *`
	`*  *`
	`*  *`
	`*  *`
	`*  *`
	`*  *`
	`* *`
	`*`
		(5+5)	

Q7	**a)** What is the output of the following C code? Assume that the address of x is 1000 (in decimal) and an integer requires four bytes of memory.
	`int main()`
	`{`
	`unsigned int x[4][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}, {10, 11, 12}};`
	`printf("%u,%u, %u", x+3, *(x+3),*(x+2)+3);`
	`}`
	b) What are the differences between while and do while loops? Explain with the help of examples.
		(5+5)	

	UNIT-IV		
Q8	Explain the following library functions using an example i) strcpy (), ii) strrev() iii) strcat() iv strlen()	**(2.5*4)**	
Q9	a) Write a program to take the input and print the record of 5 patients using structures, where the record contains the patient's name, age, weight (in integer), and height (in integer). b) Difference between union and structure with the help of example.	**(5+5)**	

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